DSA ki Baithak

CODE BAITHAK

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Recap (Previous Lecture Connection):

* Operators & Operands
* Different categories of operators
* Short - Circuit Evaluation
* Precedence & Associativity
* Expression Evaluation
* Interview Questions & Puzzles

Why Conditional Statements ?

(statements used for decision-making)

Because many problems require decision-making

Conditional Statements :

* if
* if-else
* else-if ladder

Ternary Operator (Conditional Operator)

It is a shorthand way of writing simple if-else statement

Syntax :

False

(Condition) ? expression 1 : expression 2

True

① WAP to implement the "Clamp Function".

(Clamp Function restricts value within range)

$\boxed{\text{clamp}(x, \text{min}, \text{max})}$

* clamp(12, 5, 10) → 10
* clamp(7, 5, 10) → 7
* clamp(3, 5, 10) → 5

② WAP to find the maximum of 3 numbers.

Why Loops ?

Because many problems require repeating operations.

while-Loop → Generally when no. of iterations is not Known

for-Loop → Generally used when no. of iterations is Known in advance.

Every loop consists of 3 parts -

* Initialization
* Condition
* Updation

if Condition never becomes false

↓

Infinite loop

③ WAP to find the sum of 'n' natural numbers.

④ WAP to print the multiplication table of a number.

break & continue Keywords

break → Terminates the current loop immediately.

continue → Skips the current iteration of the loop.

Scope of a Variable

Scope defines where a variable is accessible.